

Exercise Set 2 — Defense Sheet

Computational Finance (WI4151), TU Delft — Prof. Fang Fang

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This set has two problems, each in its own file: - **Exercise 1 — Least-Squares Monte Carlo (LSM / Longstaff–Schwartz):** `exercise1_def (2).py` - **Exercise 2 — Finite Difference (FTCS) for European and American puts:** `exercise2_def (2).py`

Line numbers below refer to those two files. Where a “why” requires my own reasoning, it is left as a *TODO(Bruno)* for me to answer out loud.

1 The task (in my own words)

Exercise 1 — LSM (Longstaff–Schwartz, 2001). - Part (i): reproduce the textbook 8-path numerical example from the paper. Strike $K = 1.10$, $r = 6$, three yearly exercise dates ($dt = 1$, $T = 3$). Print the cash-flow matrix at each time, the in-the-money regression tables, the fitted $E[Y|X] = a_0 + a_1X + a_2X^2$, the optimal early-exercise decisions, the stopping rule, the option cash-flow matrix, and finally the American and European put prices to 5 decimals. - Part (ii): replace the 8 hard-coded paths with $N = 100,000$ simulated GBM paths (50k + 50k antithetic), $K = 40$, $M = 50$ exercise dates per year, and replicate the “red box” columns of Table 1 of the paper (Simulated American, s.e., closed-form European, early-exercise value) for the 20 (S_0, σ, T) combinations. Here the basis is the first three weighted Laguerre polynomials evaluated at $\mathbf{x} = S/K$.

Exercise 2 — Finite Difference (FTCS). - Part (i): adapt a provided `ftcs_bs_put.py` to price European puts via an explicit (forward-time, central-space) scheme in log-price space $m = \log S$, for the same 20 parameter combinations, and compare against closed-form Black–Scholes. - Part (ii): turn the

European FTCS solver into an American one by adding the early-exercise (free-boundary) projection $U = \max(U, \text{intrinsic})$ after each time step. Compare FD-American against the LSM-American of Exercise 1 as a correctness check. - Part (iii): assemble the full “outside the red box” columns of Table 1 (FD American, closed-form European, FD early-exercise value, and the FD-minus-LSM difference in early-exercise value).

TODO (Bruno): In one sentence, why does the American put price exceed the European put price here, and why is that *not* true for an American call on a non-dividend stock?

2 My approach and why

Exercise 1 — LSM. - Backward induction over exercise dates. At each date, regress discounted realized future cash flows Y on a basis in the current spot X , but **only over in-the-money paths**, then compare immediate exercise against the fitted continuation value. - Part (i) uses a degree-2 monomial basis $1, X, X^2$ via an sklearn pipeline (`exercise1_def (2).py:107`). - Part (ii) uses the first three weighted Laguerre polynomials in $x = S/K$ (`exercise1_def (2).py:185-191`). - Variance reduction by antithetic paths (`exercise1_def (2).py:177-178`).

TODO (Bruno): Why regress only on in-the-money paths rather than all paths? (Out-of-the-money continuation values are irrelevant to the exercise decision — but say *why* in your own words and tie it to the paper.)

TODO (Bruno): Why monomials $1, X, X^2$ in Part (i) but weighted Laguerre in Part (ii)? Was this just “match the paper,” or numerical conditioning? Be ready to defend both.

TODO (Bruno): Antithetic variates: why does pairing Z with $-Z$ reduce variance for this payoff, and what property of the estimator makes the variance reduction work? State the covariance argument.

Exercise 2 — FTCS. - Work in log-price $m = \log S$ so the Black–Scholes PDE has constant coefficients, build the explicit evolution matrix F once per parameter set, and march forward in transformed time (time-to-maturity). - American: project onto the payoff after each step (`exercise2_def (2).py:105`).

TODO (Bruno): Why transform to log-price $m = \log S$ before discretizing instead of working directly in S ? (Constant coefficients / uniform grid — but explain the consequence for the matrices $T1, T2$.)

TODO (Bruno): FTCS is *explicit*. Why is that an acceptable choice here, and what stability condition are you implicitly relying on with $Nt = 40000 * T$? (See “hardest question” below — this is the danger zone.)

3 Key code sections (file + what it does + why I wrote it that way)

3.1 Exercise 1 — `exercise1_def (2).py`

- **:11-24** — Part (i) parameters and the 8 hard-coded paths from the paper. C (cash-flow matrix) initialized to zeros at **:27**, shape $(8, 3)$.
- **:32-86** — three printing helpers (`print_formatted_table`, `print_regression_table`, `optimal_table`) that produce the per-time tables shown in the report. Pure presentation, no pricing logic.
- **:89** $LSQM(S, C, dt, T, K, r)$ — the Part (i) LSM engine.

- :91 terminal payoff $C[:, T - 1] = \max(K - S[:, T], 0)$.
- :93 *C_european* keeps a copy of the terminal payoff for the European price later.
- :96 backward loop $i = T - 2 \dots 0$.
- :98 regressor $X = S[:, i + 1]$ masked by $< K$ (so OTM entries are zeroed before the ITM filter), reshaped to 2D.
- :99-101 builds $Y = \text{sum of all future cash flows discounted back to time } i + 1 \text{ using } discount_factors = \exp(-r * dt * (u - i)) \text{ with } u = \text{arange}(i + 1, n_cols)$.
- :102-104 ITM mask $S[:, i + 1] < K$; restrict X, Y to ITM.
- :107-108 degree-2 polynomial + linear regression pipeline, fit on ITM data.
- :111-112 extracts intercept $a0$ and coefficients $a1, a2$.
- :117-120 continuation = model prediction on ITM paths; exercise = $K - S$ on ITM paths.
- :122-127 exercise rule: if *exercise* > *continuation*, set $C[j, i]$ to exercise value and **zero out all later cash flows** $C[j, i + 1 :] = 0$; else $C[j, i] = 0$.
- :131 stopping matrix from $C! = 0$.
- :147-152 American price: discount each path's cash flows by $\exp(-r * dt * t)$ for $t = 1..T$, sum per path, average.
- :156 European price: $\text{mean}(C_european * \exp(-r * dt * T))$.
- :167-173 — Part (ii) parameters ($K = 40, M = 50, N = 100000$).
- :175-183 *simulate_paths* — GBM exact-step simulation; :177 draws $N//2$ normals, :178 stacks $-Z$ for antithetics; :181-182 exact log-Euler update $S[:, i + 1] = S[:, i] * \exp((r - 0.5\sigma^2)dt + \sigma\text{sigmasqrt}(dt)Z)$.
- :185-191 *laguerrebasis* — $x=S/K; L0=\exp(-x/2), L1=\exp(-x/2)(1-x), L2=\exp(-x/2)(1-2x+x^2/2)$; stacked column-wise.
- :193-196 **black_scholes_put** — closed-form European put.
- :198-235 *LSM_pricing* — Part (ii) LSM engine; same backward structure as *LSQM* but basis is *laguerrebasis* (:212) and *LinearRegression()* with default intercept (:215). Returns per-path discounted cash flows.
- :237-267 — loop over the 20 $(S0, \sigma, T)$ combos: $M_{total} = M * T, dt = 1/M$, simulate, price, compute *american* = $\text{mean}(\text{cashflow})$, *european* = **black_scholes_put(...)**, *ee* = $|\text{american} - \text{european}|$ (:257), *se* = $\text{std}(\text{cashflow})/\text{sqrt}(N)$ (:258).

TODO (Bruno): :91 and :203 index $S[:, T] / S[:, M_{total}]$ for the terminal payoff while the cash-flow matrix C has only T (resp. M_{total}) columns indexed $0..T - 1$. Explain the off-by-one convention: S has one more column (the $t = 0$ spot) than C . Be ready to point at exactly which column of S is “today” and which is maturity.

TODO (Bruno): :257 uses *ee* = $\text{np.abs}(\text{american} - \text{european})$ (absolute value) in the Part (ii) block of *exercise1_def*, but the copy in *exercise2_def* (2).py:223 uses the *signed ee* = $\text{american_value} - \text{european_value}$. Which one matches the paper’s “early exercise value” column, and is the absolute value ever wrong (can LSM-American dip below European by noise)?

TODO (Bruno): Standard error *se* = $\text{std}(\text{cashflow})/\text{sqrt}(N)$ at :258. With antithetic sampling the N draws are **not independent** (each path is paired with its antithetic). Is dividing by $\text{sqrt}(N)$ the correct s.e., or does it understate/overstate it? Defend your formula.

3.2 Exercise 2 — *exercise2_def* (2).py

- :5-14 — parameters $K = 40, r = 0.06$, and the 20 $(S0, \sigma, T)$ combos.
- :18-64 — **Part (i) European FTCS** loop:
 - :24-26 closed-form BS put for reference.

- :29 domain length $L = 2 * \log(K) + 2$; :30 $Nx = 1000$; :31 $Nt = 40000 * T$; :32-33 steps $h = L/Nx$, $k = T/Nt$.
- :35-36 central-difference matrices: $T1$ (first derivative, +1/-1 off-diagonals), $T2$ (second derivative, -2 on diagonal, 1 off-diagonals), size $(Nx - 1)$.
- :38-40 explicit evolution matrix $F = (1 - r k) I + 0.5 k \sigma^2 / h^2 * T2 + k (r - 0.5 \sigma^2) / (2h) * T1$.
- :42-43 grid $mvec$, centered and shifted by $+\log(K)$.
- :45-46 initial condition $U[:, 0] = \max(K - \exp(mvec), 0)$ (payoff as the time-0 condition in time-to-maturity).
- :48-52 forward march; :51 boundary correction term $p[0]$ injected at the low- m (deep ITM) boundary node, scaled by $K * \exp(-r * time2mat)$.
- :54 `np.interp` to read off the price at $\log(S0)$.
- :66-118 — **Part (ii) American FTCS** loop: identical to Part (i) except $intrinsic = \max(K - \exp(mvec), 0)$ (:95) and the single added projection line :105 `U[:, i+1] = np.maximum(U[:, i+1], intrinsic)`. `ee = ftcs_american - bs_put` (:108).
- :120-204 — re-imports and re-runs the **Exercise 1 Part (ii) LSM** code (paths, Laguerre basis, BS put, $LSM_{pricing}$) to regenerate `resultsismc` for the cross-check.
- :235-248 — correctness test table: $|FDAmerican - LSMAmerican|$ per row.
- :250-261 — full “outside the red box” table including the signed $FDEE - LSMEE$ difference.

TODO (Bruno): :29 $L = 2 * \log(K) + 2$. Why this width, and why center the grid at $\log(K)$ (:43)? What goes wrong at short maturity / high vol if the domain is too narrow — and what is the analogy with the COS truncation-range $[a, b]$ issue I hit in Assignment 2?

TODO (Bruno): :51 the boundary term $p[0]$. Explain in words what Dirichlet boundary condition this enforces at the deep-in-the-money edge, why it carries the factor $K * \exp(-r * time2mat)$, and what happens at the *other* (high- S) boundary where no correction is added.

TODO (Bruno): :105 is described in the report as “the single line that changes.” Explain why projecting $U = \max(U, intrinsic)$ after each explicit step is a valid (operator-splitting / explicit-penalty) treatment of the American free boundary, and what its accuracy cost is versus a proper LCP/PSOR solve.

4 Design decisions I must justify

1. **ITM-only regression** (:103-104, :211-213). Factual: the mask is $S[:, i+1] < K$. **TODO (Bruno):** justify why this is the Longstaff-Schwartz prescription and not a shortcut.
2. **Basis choice:** monomials degree 2 in Part (i) (:107) vs weighted Laguerre in Part (ii) (:185-191). Factual: the report says normalization $x=S/K$ avoids underflow in $\exp(-x/2)$ for S in $[36, 44]$. **TODO (Bruno):** confirm you can derive the underflow argument and say why monomials were fine for the small $K = 1.10$ example.
3. **Antithetic variates** (:177-178). **TODO (Bruno):** justify and quantify expected variance reduction.
4. **Exact GBM step** (:182) rather than Euler. **TODO (Bruno):** why is the log update exact for GBM, and does discretization bias still enter the American price through the discrete exercise dates regardless?
5. $M = 50$ **exercise dates/year as a proxy for continuous American exercise** (:172, :251). **TODO (Bruno):** why is a Bermudan with 50/100 dates a good stand-in for the true American, and which direction is the bias?

6. **Explicit FTCS with $Nt = 40000 * T$ (exercise2_def (2).py:31). TODO (Bruno):** show that this choice keeps the scheme inside the explicit stability region (compute the CFL-type bound; see hardest question).
7. **np.interp to extract the price at $\log(S_0)$ (:54, :107).** Factual: linear interpolation between grid nodes. **TODO (Bruno):** is linear interpolation accurate enough given h , or should it be higher order?

5 Results and what they mean

Exercise 1, Part (i) (from the report): American put price **0.11443**, European put price **0.05638** on the same 8 paths. The fitted regressions match the paper: at $t = 2$, $E[Y|X] = -1.06999 + 2.98341X - 1.81358X^2$; at $t = 1$, $E[Y|X] = 2.03751 - 3.33544X + 1.35646X^2$. Stopping rule and option cash-flow matrix reproduce the paper's table.

Exercise 1, Part (ii) (report Fig. 13): the 20-row replication of the red-box columns; e.g. $S = 36$, $\sigma = 0.2$, $T = 1$ gives Simulated American **4.4714** (s.e. **0.0094**), closed-form European **3.8443**, early-exercise value **0.6271**. s.e. all in the 0.006–0.023 range.

Exercise 2, Part (i) (report Fig. 14): FD-European vs BS-European agree to $\sim 1e-4$ across all 20 rows (largest $|Difference| = 0.0008$ at $S = 40$, $\sigma = 0.2$, $T = 1$).

Exercise 2, Part (ii) (report Fig. 16): FD-American prices, all above the corresponding European, with early-exercise values matching the LSM ones in sign and magnitude.

Exercise 2 cross-check (report Fig. 17): $|FDAmerican - LSMAmerican|$ is below 0.05 for all 20 rows (max ~ 0.016), and the Part (iii) $FDEE - LSMEE$ differences are small and alternate in sign.

TODO (Bruno): The FD-European prices are *systematically slightly below* BS (every *Difference* is negative, $\sim -1e-4$ to $-8e-4$). Is that a bias of the explicit scheme, the truncated domain, or the interpolation? Pick one and defend it — Fang will ask why the sign is consistent.

TODO (Bruno): The FD-American vs LSM-American differences are ~ 0.005 – 0.016 , an order of magnitude larger than the FD-European vs BS error ($\sim 1e-4$). Why is the American comparison so much looser? (Hint: LSM bias + 50-date Bermudan + regression noise vs a near-exact PDE.)

6 The hardest question Fang could ask here — and my answer

Likely hardest: “Your FTCS scheme is explicit. Is it stable for $Nx = 1000$, $Nt = 40000 * T$? Derive the condition and check it for your worst-case $\sigma = 0.4$.”

Factual setup from the code: in log-space the diffusion coefficient is $0.5 * \sigma^2$, $h = L/Nx$ with $L = 2\log K + 2 = 2\log 40 + 2 \approx 9.376$, so $h \approx 0.00938$; $k = T/Nt = 1/40000 = 2.5e-5$ per unit time.

The explicit-scheme stability constraint for the diffusion part is roughly $0.5 * \sigma^2 * k / h^2 \leq 1/2$, i.e. $k \leq h^2 / \sigma^2$.

TODO (Bruno): Plug in the numbers: $h^2 / \sigma^2 = (0.00938)^2 / 0.16 \approx 5.5e-4$, and $k = 2.5e-5 \ll 5.5e-4$, so the diffusion number $\approx 0.045 \ll 0.5$. State this out loud, confirm the scheme is comfortably stable, and explain *why the author picked Nt so large* (was it stability, or matching the paper's accuracy?). Also be ready to say what the $(1 - rk)$ reaction term and the $T1$ advection

term contribute to the stability bound — and whether the advection (first-derivative) term can cause spurious oscillations even when the diffusion bound is met.

Backup hardest (LSM): “Is your LSM price biased high or low, and why?” TODO (Bruno): Standard result — using the *same* paths to estimate the continuation regression and to value the option introduces a look-ahead/in-sample bias in the continuation estimate, so the exercise decision uses noisy information. State the direction of the resulting price bias and how an out-of-sample / two-pass scheme would fix it.

7 “Show me in your code where you do X” — anticipated spots

- “...filter to in-the-money paths before regressing.” `exercise1_def (2).py:102-104` ($itm_indexes = S[:, i + 1] < K$), and :211-213 for Part (ii).
- “...build the discounted continuation target Y .” :99-101 (Part i) and :207-209 / :182-184 (Part ii).
- “...fit the continuation-value regression and read off the coefficients.” :107-112 (monomial pipeline) and :215-217 (Laguerre).
- “...make the exercise-vs-continue decision and zero out future cash flows.” :122-127 (Part i), :224-229 / :194-199 (Part ii).
- “...compute the American and European prices.” :147-157 (Part i American + European).
- “...generate antithetic paths.” :177-178.
- “...evaluate the weighted Laguerre basis.” :185-191.
- “...the closed-form Black-Scholes put.” :193-196 (Ex.1), `exercise2_def (2).py:24-26` and :74-76.
- “...assemble the explicit FTCS evolution matrix F .” `exercise2_def (2).py:35-40` (and :85-90).
- “...set the initial/boundary conditions of the PDE.” initial: :45-46; boundary term: :51 (and :103).
- “...the one line that makes it American.” `exercise2_def (2).py:105` (`U = np.maximum(U, intrinsic)`).
- “...read the price off the grid at S_0 .” :54 and :107 (`np.interp(np.log(S0), mvec, U[:, Nt])`).
- “...cross-check FD against LSM.” :235-248 ($|FD_{American} - LSM_{American}|$).